Alessio **Cocchieri**

About Me

I am a final-year NLP PhD candidate at the University of Bologna (available from Nov. 2026), with 10+ first-author publications at top-tier NLP venues (e.g., ACL, EMNLP). I focus on LLM evaluation, designing benchmarks that expose systematic failures in frontier models, with extensive work in healthcare NLP and direct clinician collaboration. I also have experience using LLM synthetic supervision to improve low-resource downstream tasks, particularly zero-shot NER.

Work Experience

IBM Research — **Research Scientist Internship** (PhD Intern) Dublin (IE) | March 2026 — May 2026
Natural Language Processing — LLM Factuality (*Paper under review at ICLR 2027*)

- Built an automated agentic pipeline for constructing a multimodal scientific fact-checking benchmark
- Coordinated a multi-team human annotation pipeline, overseeing annotation guidelines, quality control, and IAA
- Conducted systematic analysis exposing critical factuality fragilities in multimodal LLMs across scientific domains

IBM Research — **Research Scientist Internship** (Master's Thesis) Dublin (IE) | March 2023 — May 2023
Natural Language Processing — Named Entity Recognition

- Trained lightweight zero-shot NER models on LLM-generated synthetic annotations from large unlabeled corpora
- Produced two peer-reviewed publications: OpenBioNER (NAACL 2025) and ZeroNER (ACL 2025)
- Shipped features and fixes to the IBM zshot open-source library for zero-shot NER inference (#82, #84)

Selected Publications (* Equal Contribution) | Full List on Google Scholar

EMNLP 2026 (Main) w/ Sant'Orsola Hospital — Zangrillo*, **Cocchieri***, et al. *Paint It, Black: Evaluating LLMs for De-Identification of Multimodal Clinical Documents*
TL;DR. We introduce the first open-source, physician-validated benchmark for de-identifying multimodal clinical documents under HIPAA regulations. We show that multimodal LLMs struggle to localize sensitive information.

EMNLP 2026 (Findings) — **Cocchieri**, et al. *Lost in Choice: Evaluating LLMs Under Extreme Multiple-Choice Scaling*
TL;DR. We build a large-scale evaluation framework for LLM decision-making with up to 1000 candidate tools/options. We uncover systematic failures in real tool-selection scenarios, with accuracy collapsing at late-middle positions.

ACL 2026 (Main) — **Cocchieri**, et al. *LLMs (Almost) Never Abstain Under Medical Uncertainty*
TL;DR. We introduce MedQAbstain, a benchmark for medical abstention under uncertainty, revealing that frontier LLMs systematically overcommit, rarely abstaining even when the question itself is hidden.

EMNLP 2025 (Main) — **Cocchieri**, et al. *Can Large Language Models Win the International Mathematical Games?*
TL;DR. We introduce MathGames, a benchmark of 2,183 text and multimodal competition problems for evaluating mathematical and logical reasoning, revealing a substantial gap between frontier LLMs and human performance.

NAACL 2025 (Findings) w/ IBM Research — **Cocchieri**, et al. *OpenBioNER: Lightweight Open-Domain Biomedical Named Entity Recognition Through Entity Type Description*
TL;DR. We introduce a 110M BERT-based model trained on synthetic LLM annotations that leverages type descriptions for zero-shot Biomedical NER, outperforming GPT-4o, specialized LLMs, and GLiNER by up to 10% F1.

ACL 2024 (Main) — Frisoni*, **Cocchieri***, et al. *To Generate or to Retrieve? On the Effectiveness of Artificial Contexts for Medical Open-Domain Question Answering*
TL;DR. We introduce MedGENIE, the first *generate-then-read* framework for medical open-domain QA, demonstrating the effectiveness of generated over retrieved contexts, significantly improving LLM QA performance.

Education

Doctor of Philosophy (PhD) — **University of Bologna** Bologna (IT) | November 2023 — November 2026
Natural Language Processing | LLM Evaluation, Benchmarking, LLM distillation, Information Extraction
Master of Science (MSc) — **University of Bologna** Bologna (IT) | September 2021 — December 2023
Grade: 110 with honors / 110 | Artificial Intelligence
Bachelor of Science (BSc) — **University of Bologna** Bologna (IT) | September 2018 — July 2021
Grade: 110 with honors / 110 | Computer Science

Skills

- **Hard:** Strong: Python, PyTorch, vLLM, Benchmark Design, LLM-as-a-Judge, LLM APIs & Tool Use, PEFT (LoRA), Scikit-Learn, Git, NumPy, Pandas; Proficient: Java (OOP), Docker, MySQL, MongoDB, LangChain
- **Soft:** Communication, Problem-Solving, Teamwork, Quick Learning, Leadership, Adaptability, Public Speaking
- **Languages:** Italian (native), English (professional fluency), Spanish (basic proficiency)

Awards

Outstanding Reviewer — Recognized for review quality and rigor at ACL 2026 and NeurIPS 2026